



K-NEAREST NEIGHBORS

INSTANCE-BASED VS MODEL-BASED

- Instance-Based
 - Stores the training data and defers computation to prediction time
 - No parameters are learned; the data itself is the model
- Model-Based
 - Learns a fixed set of parameters during training, then discards the data
 - Prediction uses only the fitted parameters

HOW KNN WORKS

WHAT IS KNN?

- A supervised, instance-based model for both regression and classification
- Fast to fit and predict on moderate-sized data
- Frequently a strong baseline despite its simplicity

K-NN METHOD

PREDICTION ALGORITHM

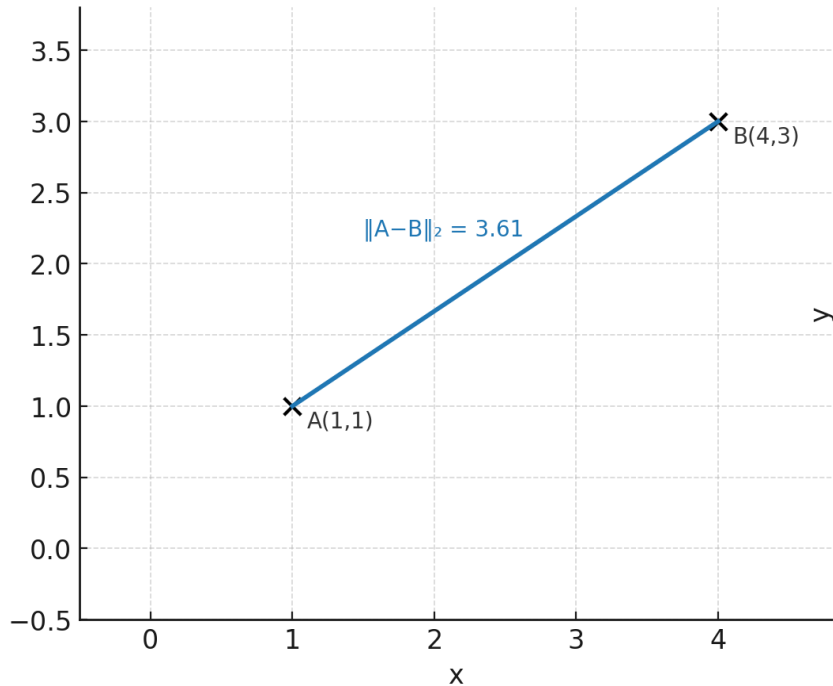
- Given a query point x , compute the distance $d(x, x_i)$ to every training point x_i , $i = 1, \dots, n$
 - How should the distance $d(\cdot, \cdot)$ be defined?
 - Let $N_k(x)$ denote the indices of the k training points nearest to x
 - Predict the target value $\hat{y}$ for the query point x :
 - Regression: $\hat{y} = (1/k) \sum y_i$, for $i \in N_k(x)$
 - Classification: $\hat{y}$ = the majority class among $\{ y_i : i \in N_k(x) \}$
 - Both are the uniform-weight case; see weighted k-NN
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CHOICE OF DISTANCE

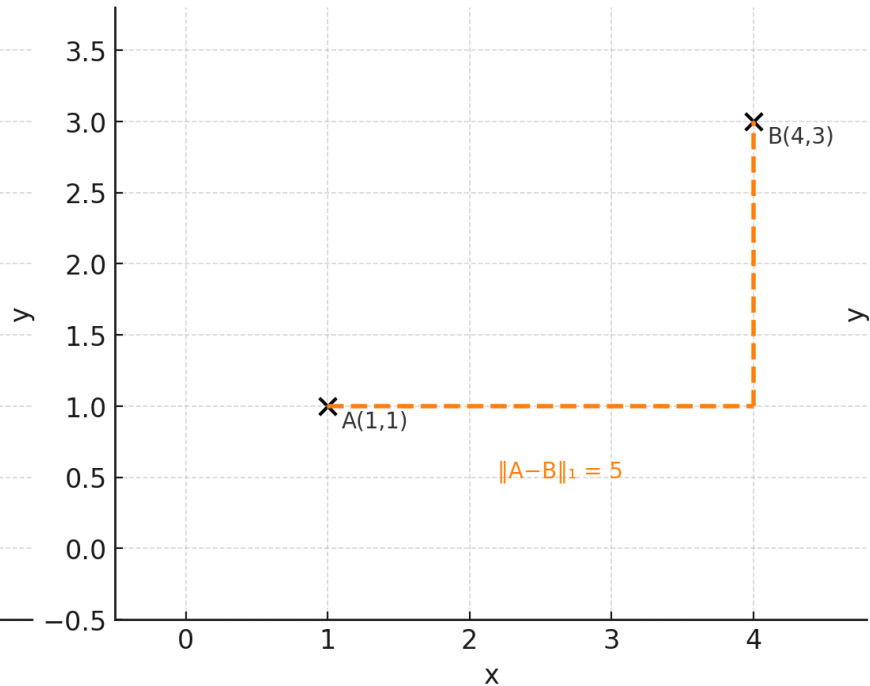
- Euclidean: the default choice for continuous, well-scaled features
- Cosine: preferred for high-dimensional or sparse data

Comparison of Euclidean, Manhattan, and Cosine Distances

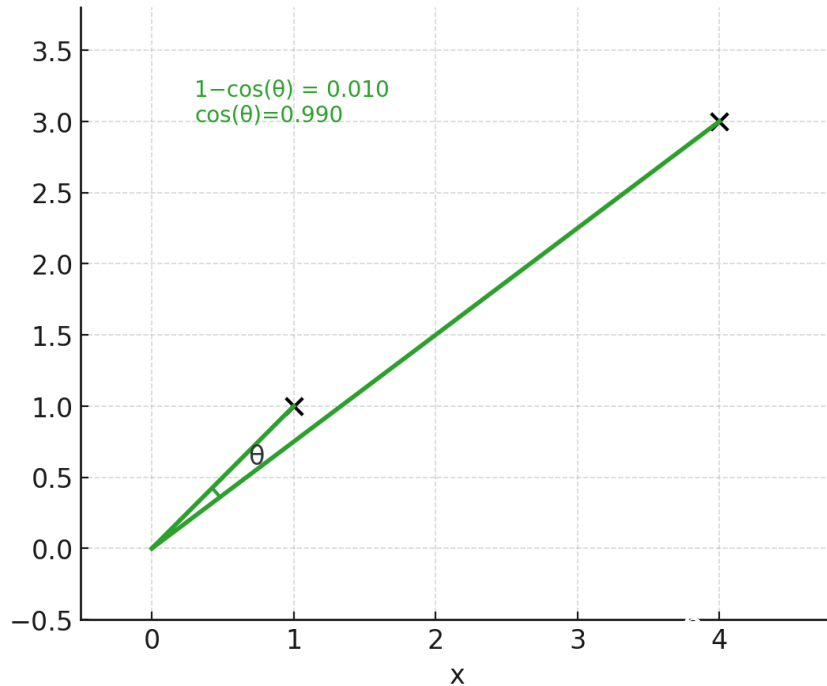
Euclidean Distance



Manhattan Distance

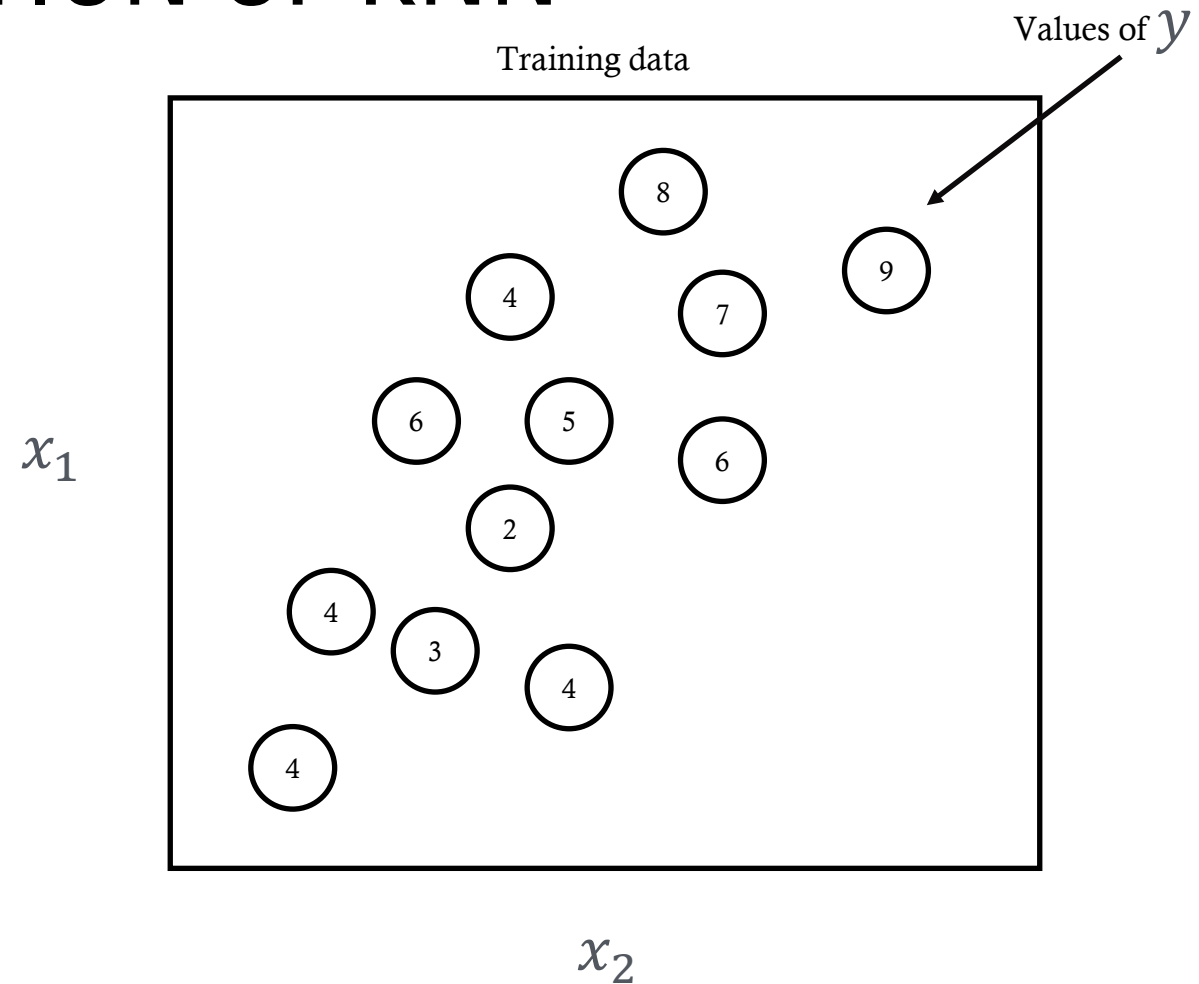


Cosine Distance

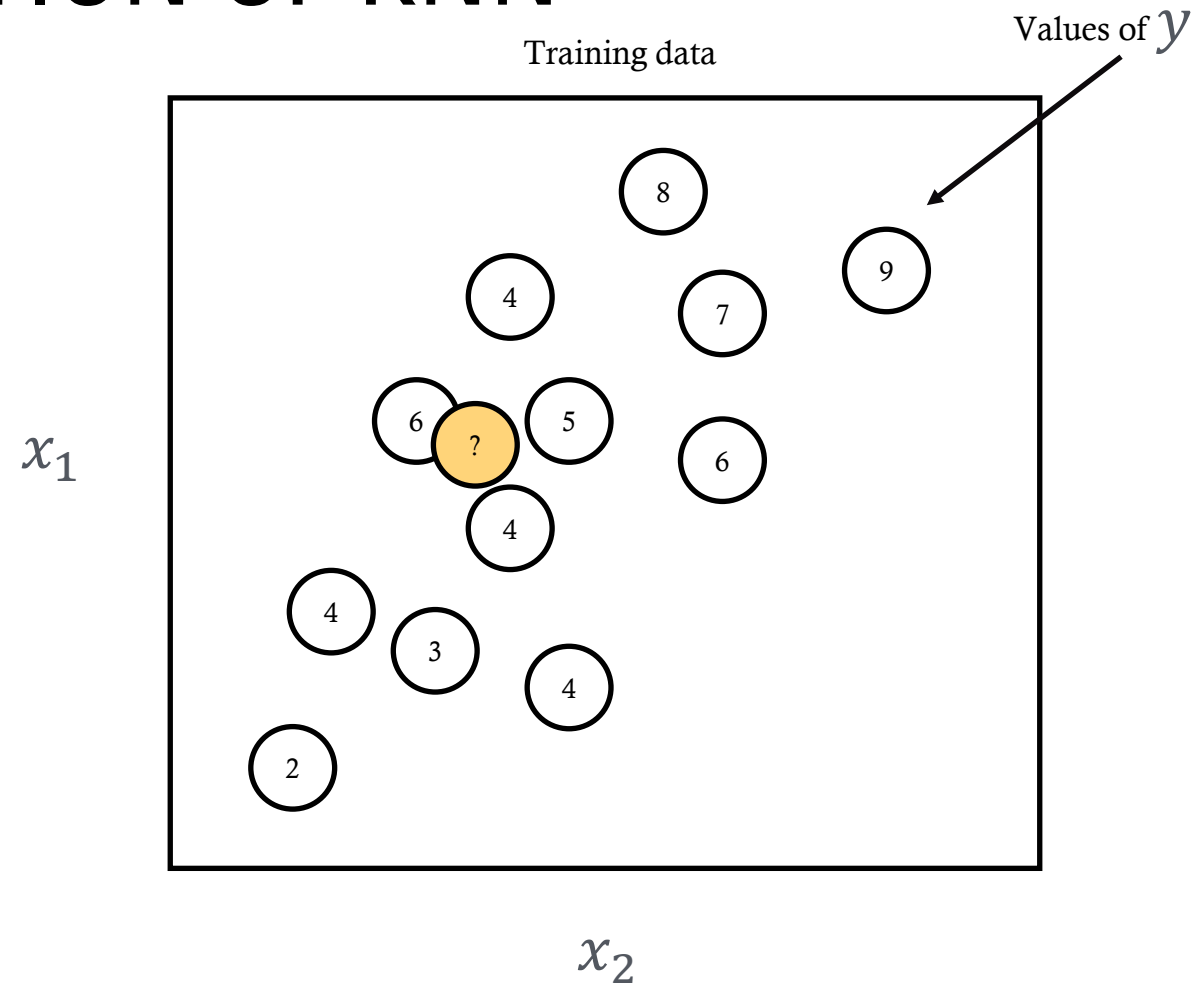


K-NN REGRESSION

VISUALIZATION OF KNN



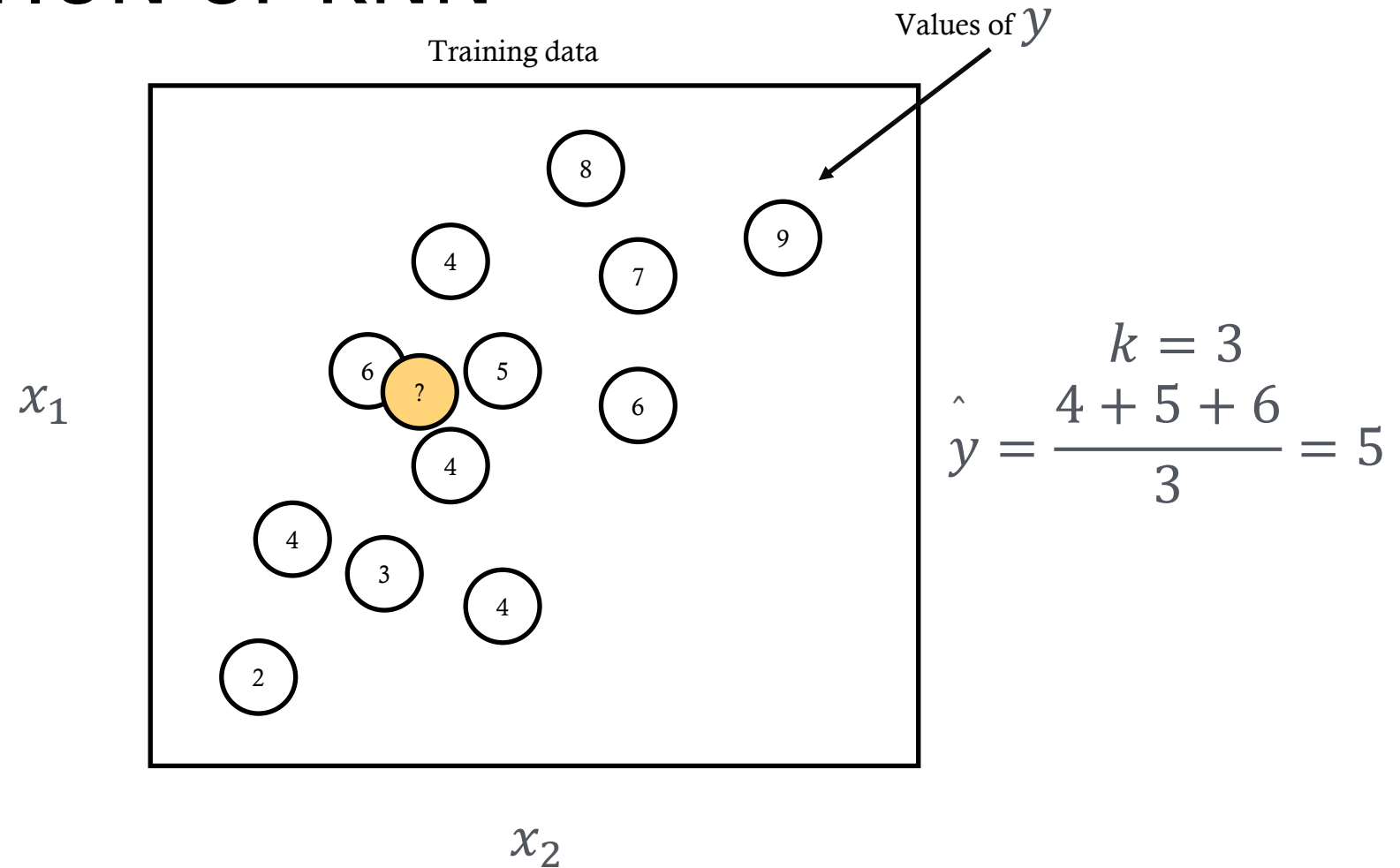
VISUALIZATION OF KNN



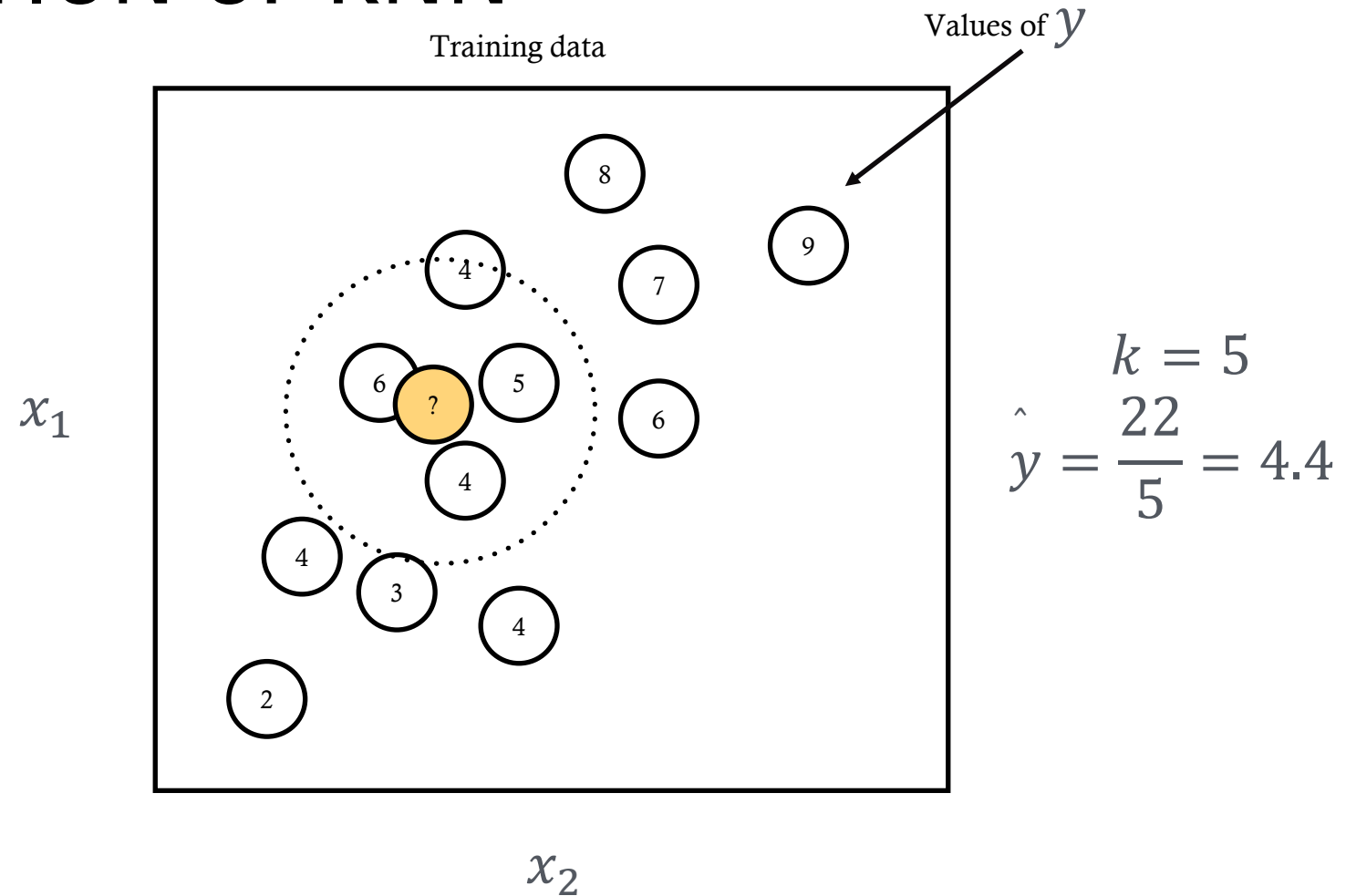
VISUALIZATION OF KNN



VISUALIZATION OF KNN

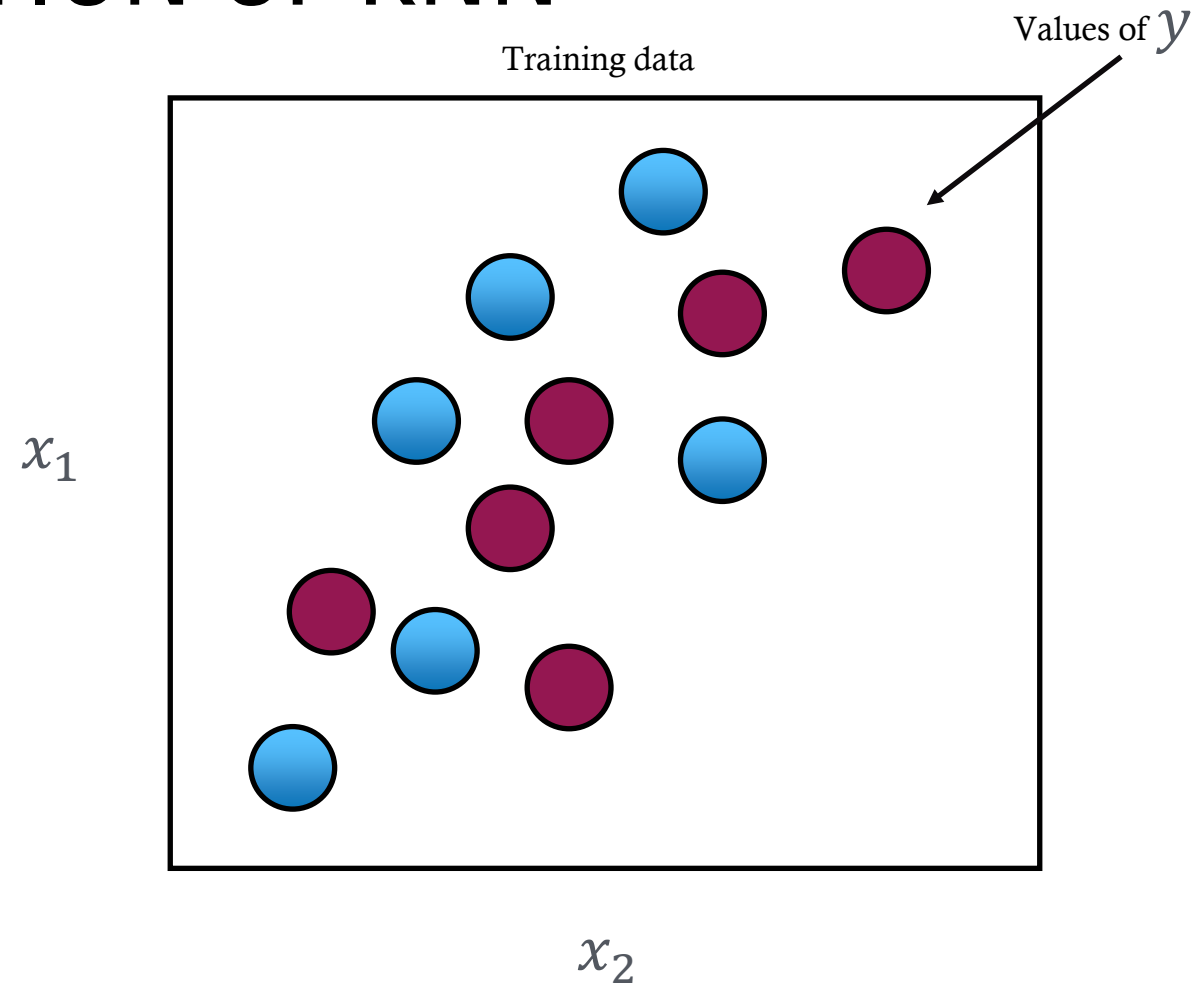


VISUALIZATION OF KNN

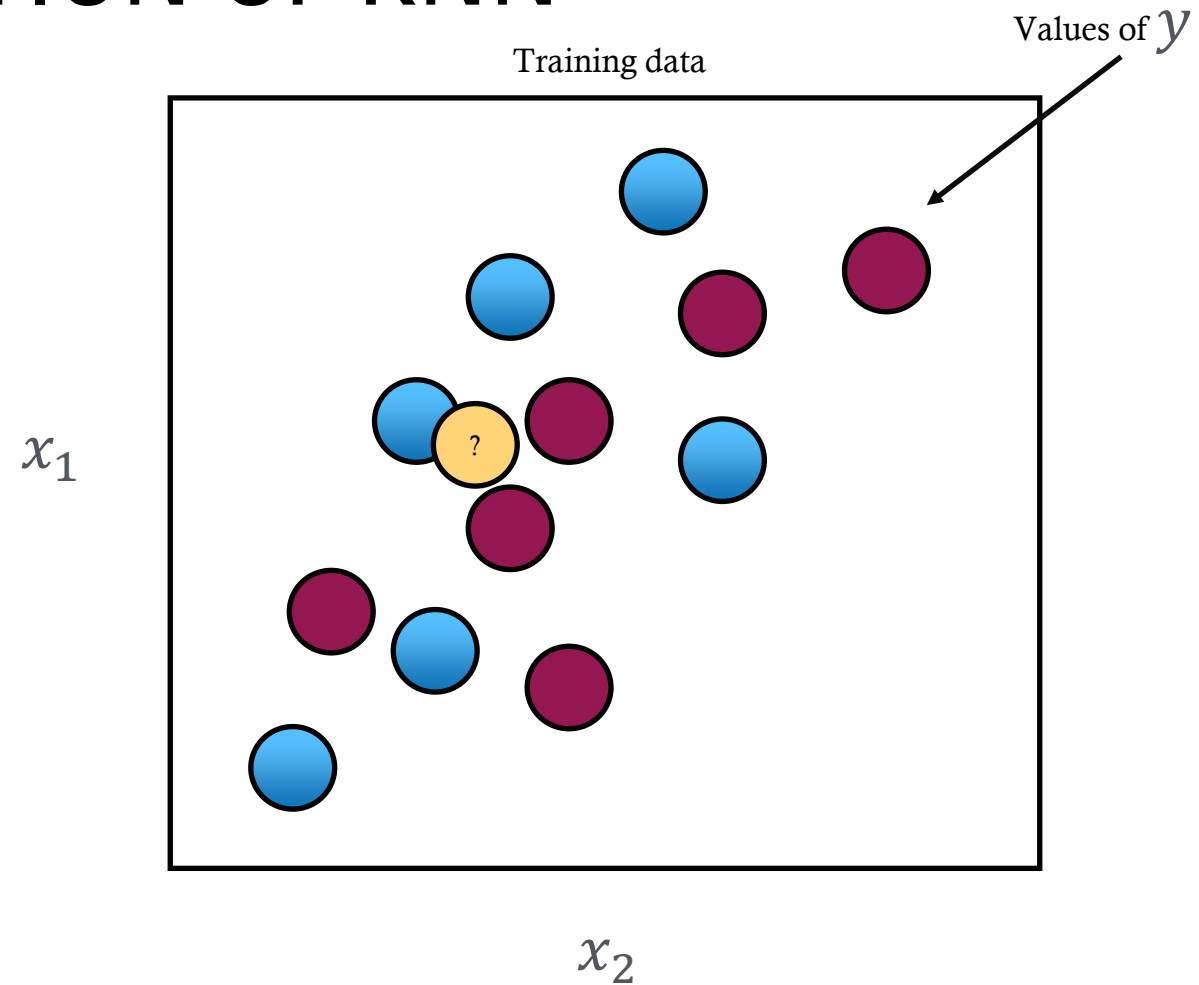


K-NN CLASSIFICATION

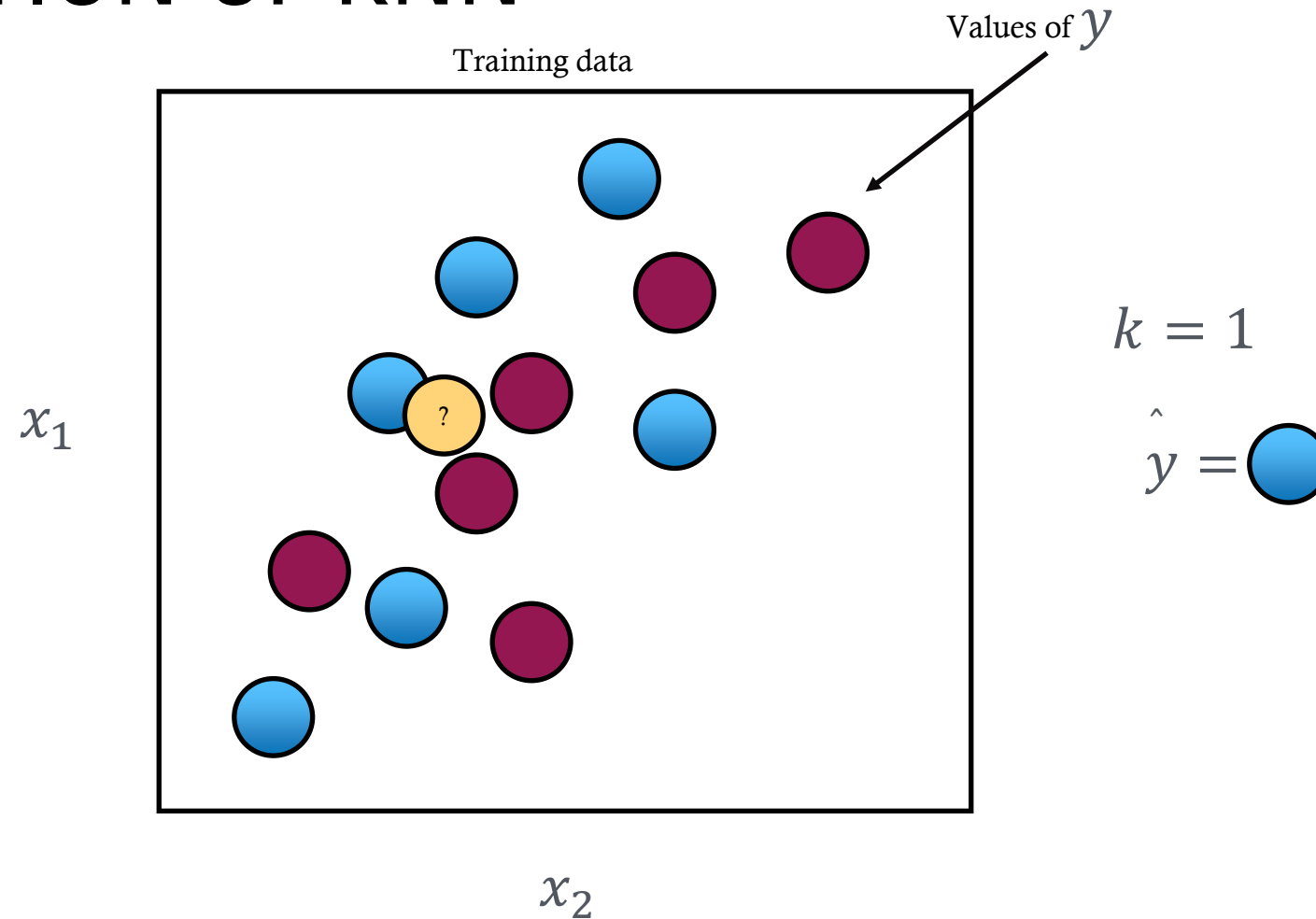
VISUALIZATION OF KNN



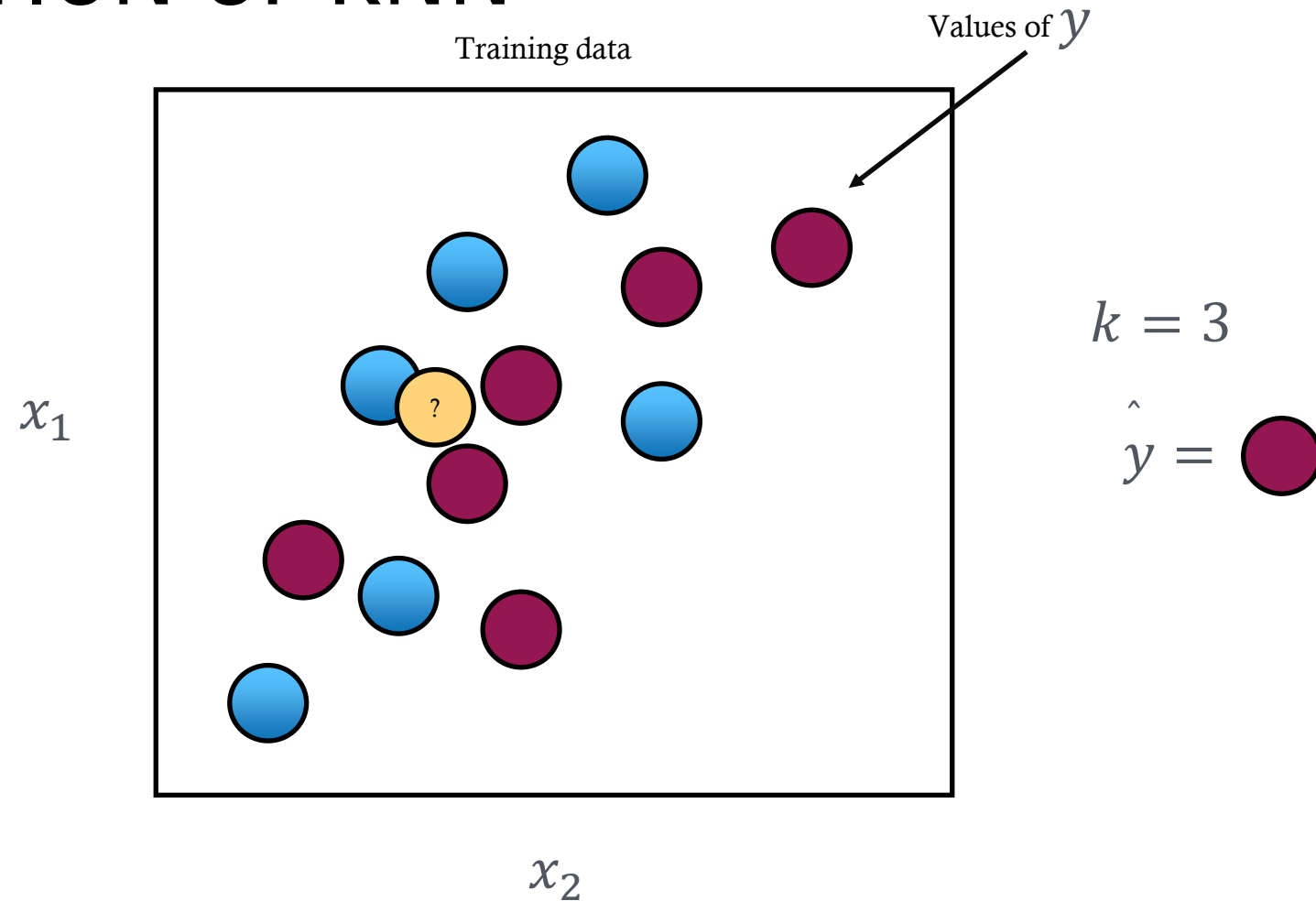
VISUALIZATION OF KNN



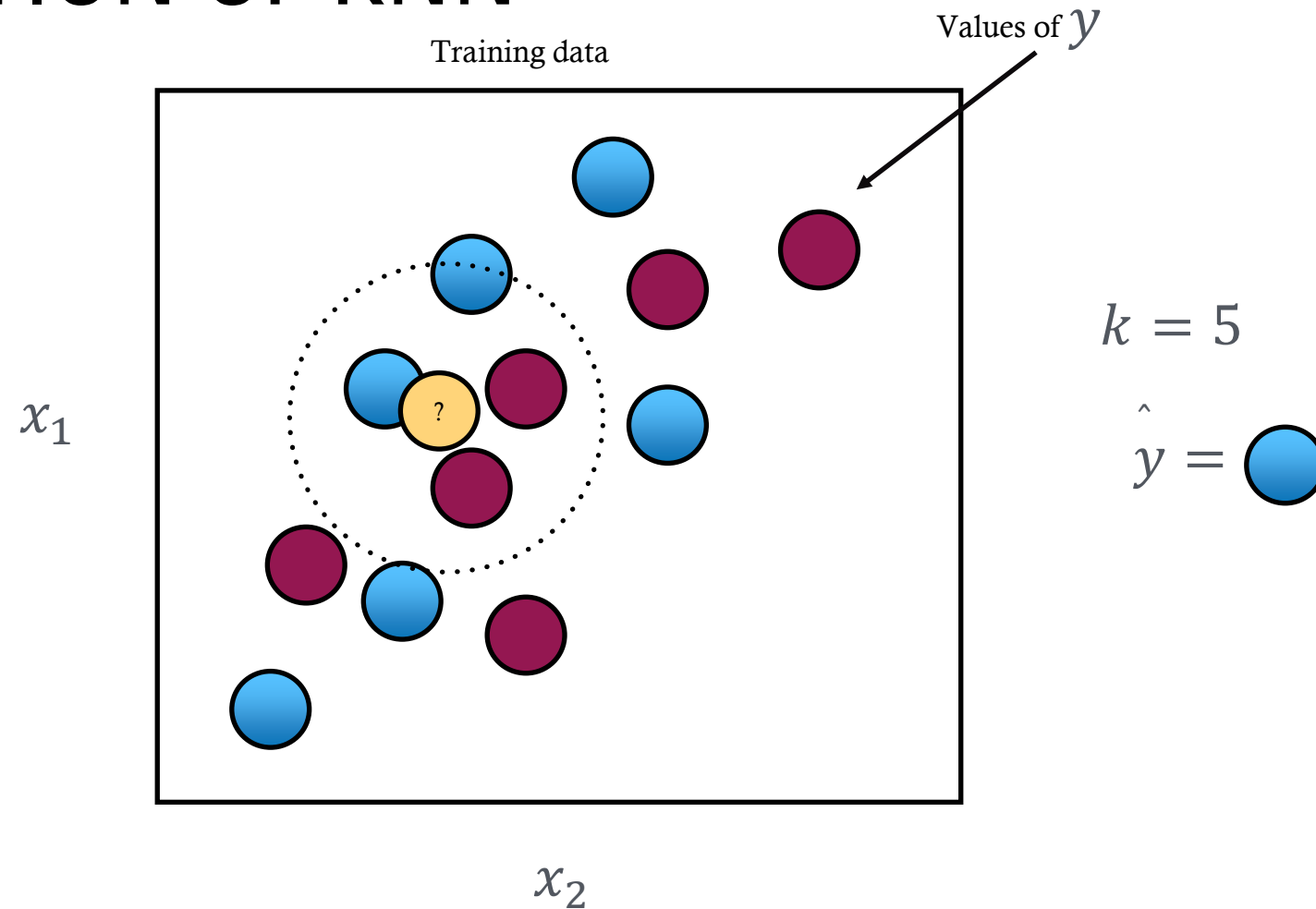
VISUALIZATION OF KNN



VISUALIZATION OF KNN

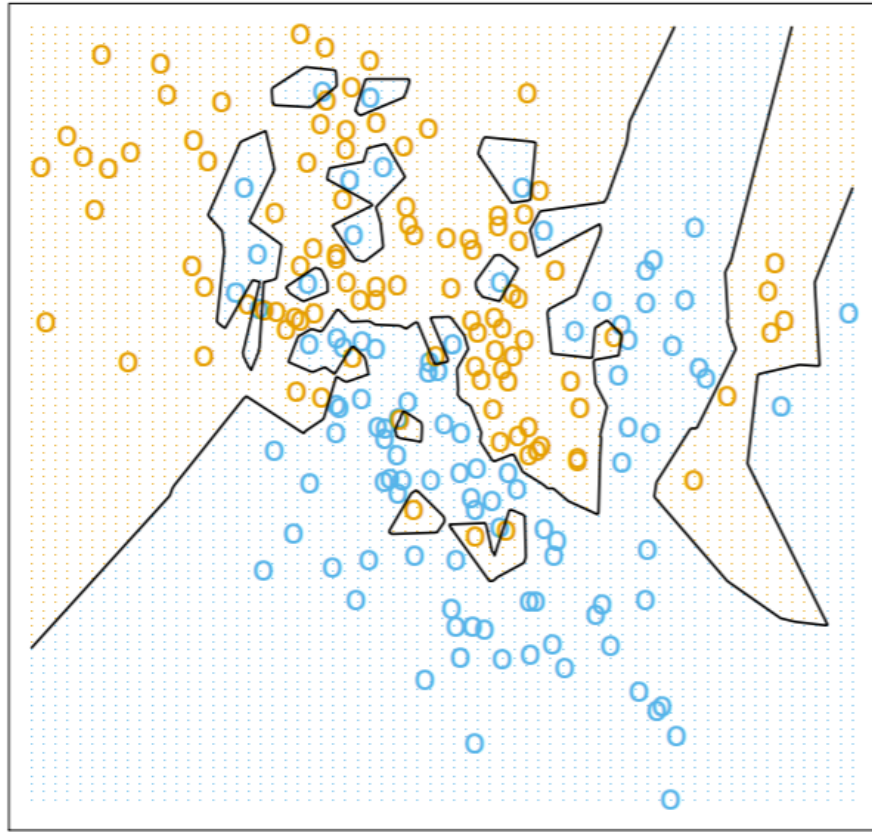


VISUALIZATION OF KNN

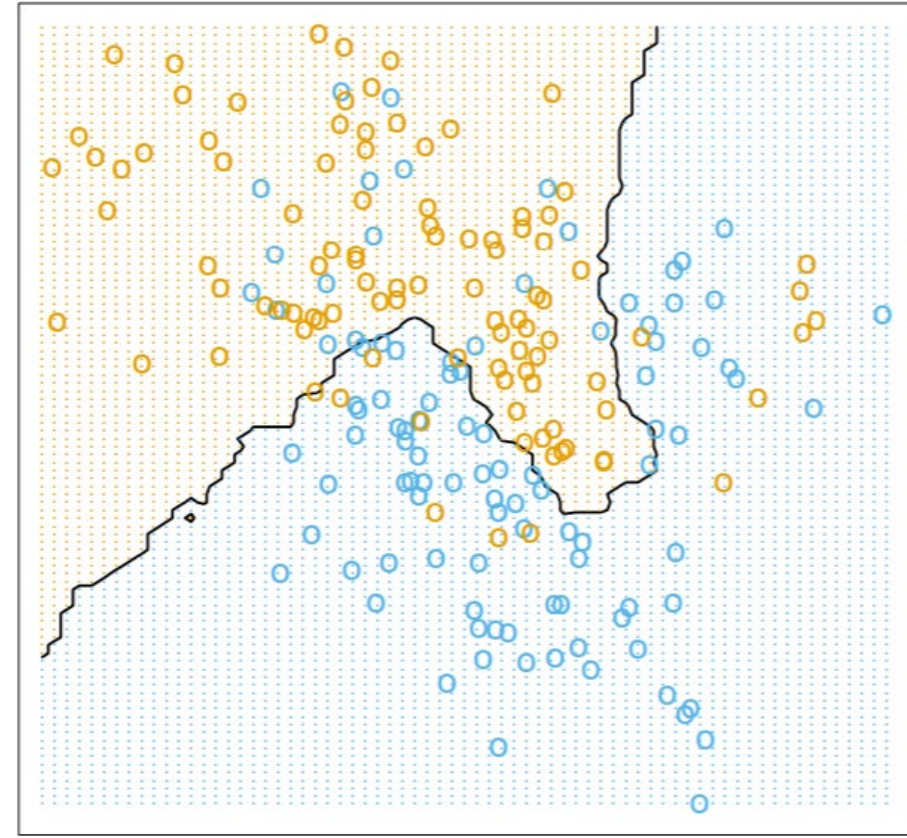


EFFECT OF K ON DECISION BOUNDARY

1 - NN



15 - NN



WEIGHTED K-NN

- Idea: weight neighbors by proximity
 - Let $w_i \propto 1 / d(x, x_i)$ for $i \in N_k(x)$
 - Closer neighbors carry more influence than distant ones
 - Regression
 - $\hat{y} = \sum w_i y_i / \sum w_i$, a weighted mean rather than a plain average
 - Classification
 - $\hat{y} = \operatorname{argmax}$ over classes c of $\sum w_i \cdot 1\{y_i = c\}$, a weighted vote
 - Why it helps
 - Larger k becomes safer: distant neighbors dilute the prediction less
 - Ties in classification break naturally
 - Caveats
 - Weighting amplifies the effect of poorly scaled features
 - $1 / d$ diverges when the query point coincides with a training point
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KNN: KEY TAKEAWAYS

- When does kNN work well?
 - Continuous, well-scaled features with moderate dimensionality
 - Simple, non-parametric, no explicit training phase
 - Choosing k
 - Small k: sensitive to noise, can overfit
 - Large k: smooths boundaries, but may underfit
 - Distance weighting reduces sensitivity to k
 - Distance metrics
 - Euclidean by default; Cosine for sparse data
 - Where does kNN struggle?
 - High dimensions: distances lose meaning, so neighbors become unreliable
 - Large datasets: distances to all points make prediction slow
 - Noisy or irrelevant features distort neighbor selection
 - Class imbalance biases predictions toward the majority class
 - Always scale features
 - Distance-based methods are dominated by large-magnitude variables
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